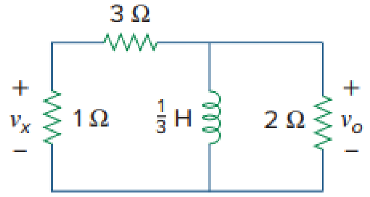
Chapter 7, Problem 23P

Problem

Consider the circuit in Fig. 7.103. Given that *vo*(0) = 10 V, find *vo* and *vx* for *t* > 0.

Fig. 7.103:



Step-by-step solution

Step 1 of 3

Refer to circuit diagram in Figure 7.103 in the textbook.

The equivalent resistance that is seen through the terminals of the inductor is both $3\ \Omega$ and $1\ \Omega$ resistors are in series and its equivalent value is in parallel with the $2\ \Omega$ resistor.

Calculate the equivalent resistance.

$$\begin{aligned} R_{eq} &= (1+3)\,||\,2 \\ &= \frac{(4)(2)}{4+2} \\ &= \frac{8}{6} \\ &= \frac{4}{3}\ \Omega \end{aligned}$$

Calculate the time constant of *RL* circuit.

$$\begin{aligned} \tau &= \frac{L}{R_{eq}} \\ &= \frac{\left(\frac{1}{3}\right)}{\left(\frac{4}{3}\right)} \\ &= \frac{1}{4}\ \text{sec} \end{aligned}$$

Step 2 of 3

Write the general expression for output voltage expression for $t > 0$.

$$v_o(t) = v_o(0)e^{-\frac{t}{\tau}}$$

Substitute 10 for $v_o(0)$ and $\frac{1}{4}$ for τ.

$$\begin{aligned} v_o(t) &= 10e^{-\frac{t}{\left(\frac{1}{4}\right)}} \\ &= 10e^{-4t} \end{aligned}$$

Therefore, the voltage $v_o(t)$ across the $2\ \Omega$ resistor for $t > 0$ is $\boxed{10e^{-4t}}$.

Step 3 of 3

Determine the voltage $v_x(t)$ for $t > 0$ using voltage division rule.

$$\begin{aligned} v_x(t) &= \left(\frac{1}{1+3}\right) v_o(t) \\ &= \left(\frac{1}{4}\right)\left(10e^{-4t}\right) \text{ V} \\ &= 2.5e^{-4t} \text{ V} \end{aligned}$$

Therefore, the voltage, v_x across the $1\ \Omega$ resistor for $t > 0$ is $\boxed{2.5e^{-4t} \text{ V}}$.